

U.S. SEMICONDUCTOR: TRADE & LABOR ANALYSIS

Occupational Wages by Role — OEWS Dataset (BLS) | 2025 Q3

Overview

This document logs the cleaning process applied to the Occupational Employment and Wage Statistics (OEWS) dataset downloaded from the U.S. Bureau of Labor Statistics. The OEWS program provides annual estimates of employment and wages for over 800 occupations across industries, geographic areas, and ownership types, making it one of the most comprehensive sources for occupational wage benchmarking in the U.S. economy.

The original dataset contained over 6 million records. The cleaning step was designed to simplify the file, remove non-analytical metadata, and ensure uniqueness — producing a leaner, audit-ready dataset prepared for large-scale labor market and wage analysis.

Source & Dataset Information

Attribute	Value
Data provider	U.S. Bureau of Labor Statistics (BLS)
Publication	Occupational Employment and Wage Statistics (OEWS)
Release	2025 (latest available at time of collection)
Format	Tab-delimited text file (.txt) — raw BLS release
Records	6,000,000+ rows across occupations, industries, and geographies

Cleaning Steps

The following steps were applied sequentially to transform the raw OEWS file into an analysis-ready dataset. Each step is documented with the corresponding code and the rationale for the decision.

Step	Action	Code	Rationale
1	Read raw dataset	<code>df = pd.read_csv(..., sep='\t')</code>	Loaded the tab-delimited OEWS file into a pandas DataFrame for inspection and processing
2	Inspect data	<code>print(df.head())</code>	Confirmed correct structure, column names, and data types before any modifications
3	Drop redundant columns	<code>df.drop(columns='footnote_codes')</code>	Removed the footnote_codes column — metadata only, no analytical value for wage or employment comparisons

4	Check for duplicates	<code>df.duplicated().sum()</code>	Verified 0 duplicate rows, ensuring accurate aggregations and preventing inflated totals
5	Save cleaned dataset	<code>df.to_csv(..., sep='\t', index=False)</code>	Exported the cleaned file in tab-delimited format, preserving the original BLS structure

Step-by-Step Detail

Step 1 — Reading the Raw Dataset

```
df = pd.read_csv('Occupational Employment and Wage Statistics (OEWS).txt', sep='\t')
```

The original OEWS dataset was loaded into a pandas DataFrame using tab-delimited parsing. The file contained millions of rows with series identifiers, employment counts, wage estimates by percentile, and metadata fields. Loading the full file confirmed the column structure prior to any modifications.

Step 2 — Inspecting the Data

```
print(df.head())
```

The first few rows were printed to confirm the dataset loaded correctly and to verify column names, data types, and value ranges before applying any transformations.

Step 3 — Dropping Redundant Columns

```
col_to_drop = 'footnote_codes'
df = df.drop(columns=col_to_drop)
```

The footnote_codes column was removed from the dataset. This field contained BLS metadata flags — indicators of data reliability, suppression, or estimation method — that add no value for wage or employment comparisons. Removing it reduces file size and eliminates noise without affecting the analytical core of the dataset.

Step 4 — Checking for Duplicates

```
print(df.duplicated().sum())
```

A duplicate check confirmed that the dataset contained 0 duplicate rows. This ensures that aggregations — such as total employment by occupation or average wages by region — produce accurate, non-inflated results. The confirmation also validates that the raw BLS file was structurally sound prior to cleaning.

Step 5 — Saving the Cleaned Dataset

```
df.to_csv('Clean Occupational Employment and Wage Statistics (OEWS).txt', sep='\t', index=False)
```

The cleaned dataset was exported in tab-delimited format to match the structure of the original BLS file. The result is a leaner file focused on the analytical essentials: occupational identifiers, employment counts, wage estimates, and wage percentiles.

Key Decisions and Rationale

- Footnote codes dropped: These were metadata only and did not contribute to wage or employment analysis. Their removal reduces file size without any loss of analytical value.
- Duplicates confirmed absent: Zero duplicate rows verified, ensuring reliable totals and averages across all downstream aggregations.
- Large-scale efficiency: Cleaning steps were designed to handle over 6 million records without sacrificing performance or auditability.
- Business focus sharpened: Retaining only occupational identifiers, employment counts, and wage measures ensures the dataset directly supports labor market analysis and wage benchmarking.
- Transparency ensured: Every cleaning step was documented with accompanying code, making the process fully reproducible and audit-ready.

How This Dataset Fits the Broader Analysis

Connection to ECI and SAE Data

Where the ECI measures the rate of change in compensation costs and the SAE tracks employment counts by state and metro area, the OEWS dataset provides the absolute wage level by occupation. Together, these three datasets form the foundation of the labor cost model: how much workers earn, how fast that cost is rising, and where workers are located.

Connection to Semiconductor Trade Analysis

Occupational wage estimates for engineering, manufacturing, and technical roles are directly relevant to semiconductor cost modeling. The OEWS dataset enables precise benchmarking of wages for roles such as semiconductor processing technicians, electrical engineers, and computer hardware engineers — the core workforce categories for U.S. fab operations.

Data Source

[U.S. Bureau of Labor Statistics — Occupational Employment and Wage Statistics \(OEWS\)](#)